KAZI TANVIR HASAN, PhD

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PROFESSIONAL SUMMARY

PhD-trained biostatistician and data scientist working across **statistical genetics, pharmacometrics, statistical methodology, causal inference, and reproducible statistical computing**. Currently Senior Associate Data Scientist, Pharmacometrics at Alnylam Pharmaceuticals. Postdoctoral research at Yale School of Medicine (Gelernter Lab) applied phenome-wide LDSC genetic correlation and Mendelian randomization (MRlap, TwoSampleMR) to 1,669 clinical phenotypes in the Million Veteran Program, with additional summary data from UK Biobank, All of Us, PGC-SUD, Yale-Penn, and 23andMe. Doctoral research developed and benchmarked Bayesian methods through large Monte Carlo studies, and produced the open-source CRAN package *simBKMRdata*. Prior industry experience at Johnson & Johnson (Clinical Pharmacology & Pharmacometrics), including basic NONMEM exposure. Advanced R, working Python, SAS, and SQL, Git, and Quarto. Five conference presentations (NIDA 2026; JSM 2022–2025) and peer-reviewed publications.

CORE COMPETENCIES

Statistical Genetics	Phenome-wide LDSC genetic correlation; Mendelian randomization (MRlap, TwoSampleMR, MR-Egger, weighted median); PRS; GWAS and PheWAS summary statistics; pleiotropy assessment; multiplicity control at biobank scale
Biobank-Scale Data	Million Veteran Program (1,669 traits, 26 medical domains); UK Biobank; All of Us; PGC-SUD; Yale-Penn; 23andMe; clinical phenotype and laboratory biomarker integration
Pharmacometrics	Pharmacometric modeling and simulation workflows; exposure-response concepts; NONMEM (basic); R-based analysis scripts and visual summaries
Statistical Methodology	Bayesian/hierarchical modeling; Monte Carlo simulation; longitudinal/mixed-effects models (LMM/GLMM); regression/GLM; causal inference; experimental design principles; sensitivity and robustness assessment
Statistical Computing	R (advanced); Python (working); SAS (working); SQL; Git/GitHub; Quarto/R Markdown; reusable functions, automated reporting, CRAN package development
Communication & Training	Presentations at JSM and NIDA; manuscripts; 10-week national R training series; explaining statistical concepts to non-statisticians; mentoring junior analysts

PROFESSIONAL EXPERIENCE

Senior Associate Scientist, Pharmacometrics | Alnylam Pharmaceuticals
Cambridge, MA · Aug 2026 – Present

- Developing reproducible analytical workflows in R supporting quantitative drug-development projects in a regulated rare disease environment.
- Collaborating with Clinical Pharmacology, Pharmacometrics, and Data Science colleagues on modeling and simulation activities.

Postdoctoral Associate, Department of Psychiatry | Yale School of Medicine (Gelernter Lab)
New Haven, CT · Jul 2025 – Sep 2026

- Built a two-stage pipeline on the Million Veteran Program: phenome-wide LDSC genetic correlation of five substance use disorders against 1,669 phenotypes in 26 medical domains, followed by MRlap Mendelian randomization to correct for sample overlap, weak instruments, and winner's curse.
- Assessed pleiotropy and robustness with MR-Egger and weighted median estimators; applied PRS; integrated GWAS summary statistics from PGC-SUD, UK Biobank, All of Us, Yale-Penn, and 23andMe in version-controlled R workflows (LDSC, TwoSampleMR, MRlap, GenomicSEM).
- Presented as first author at the NIDA Genetics & Epigenetics Cross-Cutting Research Team Meeting (June 2026); manuscript in preparation with Joel Gelernter, PhD and Joseph D. Deak, PhD.

Summer Intern — Clinical Pharmacology & Pharmacometrics | Johnson & Johnson
Spring House, PA · May – Aug 2024

- Developed reproducible R-based analytical scripts, simulations, and visual summaries supporting modeling and simulation work in a pharmaceutical drug-development setting; basic exposure to NONMEM.
- Communicated modeling assumptions and results to cross-functional scientific collaborators; conducted literature reviews on pharmacokinetics, oncology biomarkers, and quantitative drug-development methods.

Graduate Assistant, Department of Biostatistics | Florida International University
Miami, FL · Aug 2021 – Aug 2025

- Designed Monte Carlo studies of Bayesian kernel machine regression under varying sample size, exposure distribution, skewness, variability, and correlation; developed an adaptive variable-selection threshold method and benchmarked it by simulation.
- Released the open-source CRAN package *simBKMRdata*; wrote reusable R functions, simulation frameworks, and reporting templates.

- Provided study-design consultation and regression, multivariate, and diagnostic analyses for biomedical collaborators; co-authored peer-reviewed publications; presented at four consecutive JSMs (2022–2025).

Instructor Assistant, RCMI | Morehouse School of Medicine
Atlanta, GA · Feb – May 2023

- Delivered a 10-week national R programming and reproducible-analysis training series based on a modified Data Carpentry curriculum; developed scripts, exercises, and teaching materials.

Research Assistant, Department of Disaster Science | University of Dhaka
Dhaka, Bangladesh · Jan – Jul 2021

- Integrated heterogeneous datasets and built statistical summaries, visualizations, and dashboards for public-health research.

Graduate Teaching Assistant, Department of Mathematics | Illinois State University
Normal, IL · Aug 2018 – May 2020

- Developed instructional resources, facilitated study sessions, and graded assignments.

EDUCATION

PhD in Public Health (Biostatistics and Data Analytics) *Aug 2025*
Florida International University, Miami, FL

MS in Mathematics (Applied Statistics) *May 2020*
Illinois State University, Normal, IL

BS (Hons) in Statistics *Jul 2017*
University of Dhaka, Dhaka, Bangladesh

PUBLICATIONS

1. **Tanvir Hasan, K.**, Odom, G., Bursac, Z., & Ibrahimou, B. (2025). The sensitivity of Bayesian kernel machine regression (BKMR) to data distribution: a comprehensive simulation analysis. *Journal of Statistical Computation and Simulation*, 96(7), 1752–1771. doi:10.1080/00949655.2025.2608780.
2. **Hasan, K. T.** & Akman, O. (2021). Mortality Modeling Under Stochastic Frailty. *Missouri Journal of Mathematical Sciences*, 33(1), 105–118. doi:10.35834/2021/3301105.
3. Ahamed, H., **Hasan, K. T.**, Islam, M. T., & Galib, F. C. (2020). Lockdown policy dilemma: COVID-19 pandemic versus economy and mental health. *Journal of Biomedical Analytics*, 3(2), 37–58.

PREPRINTS & IN PREPARATION

1. **Hasan, K. T.**, Gelernter, J., & Deak, J. D. (2026). Leveraging phenome-wide data in the Million Veteran Program: Genomic relationships between substance use disorders and a broad spectrum of health indicators and medical conditions. *Manuscript in preparation.*
2. **Hasan, K. T.**, Odom, G., Bursac, Z., Lucchini, R., & Ibrahimou, B. (2025). Enhancing Bayesian Kernel Machine Regression: A Dynamic Thresholding Framework to Address Variability and Skewness in High-Dimensional Environmental Health Data. *medRxiv*. doi:10.1101/2025.04.14.2532582
3. Ibrahimou, B., **Hasan, K. T.**, et al. (2024). Assessing the risk of heart attack: A Bayesian kernel machine regression analysis of heavy metal mixtures. *Research Square*. doi:10.21203/rs.3.rs-4456611/v1.

SOFTWARE

simBKMRdata (R package). Simulation utilities for multivariate skewed exposure data and methodological evaluation of Bayesian kernel machine regression. CRAN: simBKMRdata.

CONFERENCE PRESENTATIONS

1. **Hasan, K. T.**, Gelernter, J., & Deak, J. D. (2026). Leveraging phenome-wide data in the Million Veteran Program: Genomic relationships between substance use disorders and a broad spectrum of health indicators and medical conditions. *NIDA Genetics & Epigenetics Cross-Cutting Research Team Meeting*, June 4–5, 2026.
2. Hasan, K. T., Ibrahimou, B., & Odom, G. (2025). Adaptive Thresholding in Bayesian Kernel Machine Regression: Improving Sensitivity and Reliability. *JSM*, Nashville, TN.
3. Hasan, K. T., Ibrahimou, B., & Odom, G. (2024). The Sensitivity of Bayesian Kernel Machine Regression to Data Distribution. *JSM*, Portland, OR.
4. Hasan, K. T., Ibrahimou, B., & Odom, G. (2023). Impact of heavy metal mixtures on cognitive decline and Alzheimer's disease and related dementias. *JSM*, Toronto, Canada.
5. Hasan, K. T., Ibrahimou, B., & Burchfield, S. (2022). A Bayesian kernel machine regression approach to assessing the effect of heavy metal mixtures in blood and urine on the risk of heart attack. *JSM*, Washington, DC.

REFERENCES

Available upon request.